Reflection Journal Report for Lab 07

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The purpose of this report is for each student to reflect on their learning journey for ITAI 1371 Machine Learning (ML) Lab 07. This Reflection Journal Report consists of separate parts for each student in the group. Each student in the group was responsible for their own content and report formatting in each of the sections below.

This reflection journal entry is for Lab XX TITLE OF LAB and was prepared by YOUR NAME.

This reflection journal entry is for Lab 07 Model Evaluation and was prepared by Irene "Alex" Ramos:

The purpose of this report is to reflect on my learning journey for ITAI 1371 Machine Learning Lab 07, which focused on moving beyond simple accuracy metrics to implement robust model evaluation techniques using the Titanic dataset.

Understanding Insights & New Breakthroughs

A primary breakthrough in this lab was recognizing the limitations of relying solely on baseline accuracy scores. Through hands-on practice, I learned how easily an accuracy score can be misleading when class distributions are uneven, and how utilizing a confusion matrix breaks down classifier performance into True Positives, True Negatives, False Positives, and False Negatives visualized through a Seaborn heatmap.

Analyzing the Problem / Problem Solving Strategies

When working through the tasks, I had to carefully configure scikit-learn metrics and build a baseline classifier using LogisticRegression to predict passenger survival based on features like Age, Pclass, Sex, and Fare. Furthermore, implementing 5-fold cross-validation via cross_val_score provided a stable, averaged performance score (achieving an average cross-validation score of roughly 78.68% with a low standard deviation of 0.0094), which mitigated the "luck of the draw" effect from a single train-test split.

Real-World Applications

Evaluating classification models using precision, recall, and F1-score from a classification report directly translates to practical system design. In security operations and anomaly

detection, tuning a model to prioritize precision (when false positives carry high costs, such as filtering legitimate traffic or emails) versus recall (when missing a positive case carries critical risk, such as failing to flag an intrusion) prevents costly operational blind spots.

Connecting New & Prior Knowledge

This lab effectively bridged foundational binary classification concepts with rigorous model validation pipelines. Using cross-validation reinforced the necessity of testing algorithms across multiple data folds to ensure models generalize reliably to unseen data rather than memorizing a single training subset.

Emerging Questions

As I analyzed the precision-recall trade-offs across different model thresholds, it raised the question: how can we dynamically adjust classification thresholds in production systems to balance business costs when the cost of false positives versus false negatives shifts over time?

Changing Approach & Critical Thinking

Moving forward, I will always incorporate cross-validation and detailed classification reports during the evaluation phase rather than stopping at standard accuracy scores. Understanding the cost asymmetry of classification errors is essential before deploying predictive models in real-world environments.

This reflection journal entry is for Lab 07 Model Evaluation and was prepared by Yuanyuan Kou:

Firstly, I learned how to evaluate a classification model using a confusion matrix, precision, recall, F1-score, and cross-validation. The model used Titanic passenger information to predict survival. This lab helped me understand why accuracy alone cannot tell us everything about a model.

The confusion matrix made the prediction results easier to understand. The model correctly predicted 90 passengers who died and 54 passengers who survived. It also predicted survival for 15 passengers who died and predicted death for 20 passengers who survived. Looking at these numbers helped me see the two different types of mistakes.

The main challenge for me is remembering the difference between precision and recall. Both use true positives, so their formulas can look similar. I can understand them better by asking two questions. Precision asks, “Of the passengers predicted to survive, how many

actually survived?" Recall asks, "Of all the passengers who actually survived, how many did the model find?" For survival, the model had about 78% precision and 73% recall. This means it still missed some actual survivors.

I also learned why cross-validation is useful. A single train-test split may give a different score depending on how the data is divided. Five-fold cross-validation evaluates the model using five different validation groups. The average accuracy was about 78.68%. Checking several folds gives me a better picture of how consistently the model performs.

This lesson connects to our Netflix movie project. We want to predict whether a movie may be appropriate for children. If we define unsafe movies as the positive class, recall would be especially important because we do not want to miss movies that need review. This also helped me understand that I must check what the positive class means before interpreting precision and recall.

After this lab, I understand that choosing an evaluation metric depends on the goal of the project. I still need more practice reading classification reports and explaining the results in my own words. In future work, I will look at the types of mistakes a model makes and use several metrics before deciding whether it performs well.

This reflection journal entry is for Lab 07 Regression and Classification was prepared by Carmen Esono

In this lab, I learned how to evaluate a machine learning model using different techniques, including the confusion matrix, precision, recall, F1-score, and cross-validation. I used the Titanic dataset to predict whether passengers survived or died. This activity helped me understand how machine learning models make predictions and how we can measure their performance.

One of the most important things I learned is that accuracy alone is not enough to determine whether a model is good. The confusion matrix shows the correct and incorrect predictions, helping us understand the mistakes made by the model. Precision tells us how many positive predictions were correct, while recall measures how many actual positive cases the model identified. The F1-score combines precision and recall into one measurement.

I also learned about cross-validation, which tests a model using different parts of the dataset. This method is useful because it gives us a more reliable idea of how the model might perform with new data. Instead of depending on only one train-test split, we can compare the results from several tests.

Another important lesson was understanding how these metrics apply to real-world situations. For example, precision is important in email spam filtering because we do not want important messages to be incorrectly marked as spam. Recall is especially important in cybersecurity because missing a real threat could cause serious problems.

Overall, this lab helped me improve my understanding of machine learning and model evaluation. I learned that choosing the right evaluation metrics depends on the problem we are trying to solve. In the future, I will pay more attention to different performance measurements instead of looking only at accuracy. These skills will help me build and evaluate more reliable machine learning models.

This reflection journal entry is for Lab 07 Regression and Classification and was prepared by Xiomara Torres:

This lab has helped me understand the difference between a model's overall accuracy and the types of mistakes it makes. Our model correctly predicted 144 out of 179 passenger outcomes, which gives it an 80.4% accuracy score. The confusion matrix also showed 15 false positives and 20 false negatives. Those results helped me understand why we can't rely on one accuracy score to explain everything about a model's performance.

I started off preparing the data by converting the 'Sex' column into numerical values and filling missing ages with the median. I used Age, Pclass, Sex and Fare to train the logistic

regression model after splitting the dataset. I corrected formatting problems and addressed a pandas warning by assigning the filled ages directly back to the age column. Working through these issues helped me better understand the code. The classification report showed approximately 78% precision.

I was able to connect my Python practice with a complete machine learning workflow. I learned that importing libraries provides tools, while model.fit() learns patterns from training examples and model.predict() applies those patterns to other inputs. I also learned that the logistic regression model we used does not contain artificial neurons which clarified a misconception I had about machine learning.

During this lab I questioned what would happen if a dataset or library became unavailable. This led me to think about saving trained models and backing up data.

Moving forward, I will look beyond overall accuracy and examine individual errors and variation across evaluations. I also want to practice modifying and explaining the code independently so I can build confidence in both programming and model evaluation.